

Project Presentation of Company Oriented Project (COOP)
On
Security Risks in Wearable IoT Devices

Authors:

Apoorva(2210991327), Nandni Singh(2210991960)

Supervised By
Anshu Singla

Department of Computer Science and Engineering,
Chitkara University, Punjab

Research Focus: Analyzing security vulnerabilities in wearable IoT devices and proposing lightweight cryptographic solutions

What Are Wearable IoT Devices?

- Smartwatches, fitness trackers, medical wearables.
- Collect sensitive data continuously (biometric, health, location).

Research Questions:

1. What are the major security threats?
2. Why do traditional security measures fail?
3. Which lightweight solutions are effective?

Year	Devices(in Millions)	Growth(%)
2021	929	+11.3
2022	1110	+19.5
2023	1200	+8.1
2024	1300	+8.3
2025	1400	+7.7

Table 1: Growth of connected wearable devices worldwide(2021-2025)

Key Statistics:

- **107% increase** in IoT malware attacks (2024)
- **\$4.88M** average data breach cost
- **18.5 billion** IoT devices globally (2024)

Why Does This Problem Exist?

1. Technical Constraints:

Limited CPU, memory, battery life
Cannot run traditional security (AES, RSA too heavy)

2. Manufacturer Issues:

Prioritize speed-to-market over security
Cost cutting eliminates security features
Ship devices with known vulnerabilities

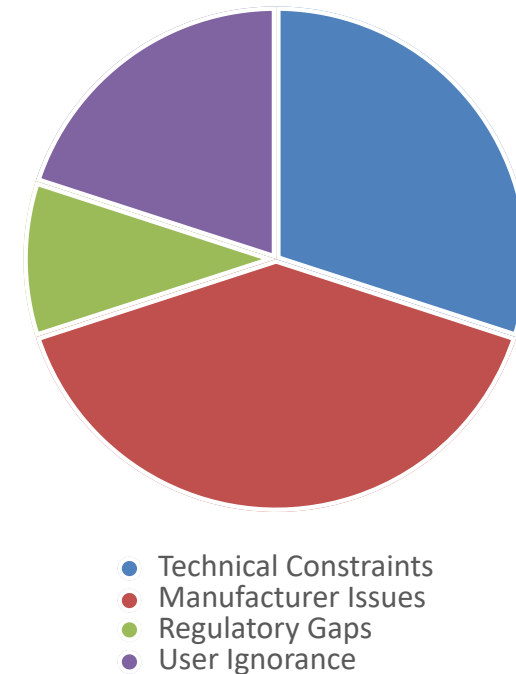
3. User Issues:

Unaware of security risks
Don't change default passwords
Don't update firmware

4. Regulatory Gaps:

No mandatory security standards
Weak enforcement

Root Cause



Market Scale & Urgency:

Critical Vulnerabilities Found:

- 60-70% have weak BLE security
- 40% use default credentials
- 30-40% have hard-coded passwords

Why Solutions Are Urgent:

Healthcare data breaches: **\$10.93M** average cost

Mirai botnet infected **600,000+** IoT devices

Patient safety at risk with medical wearables

Privacy breach of sensitive biometric data

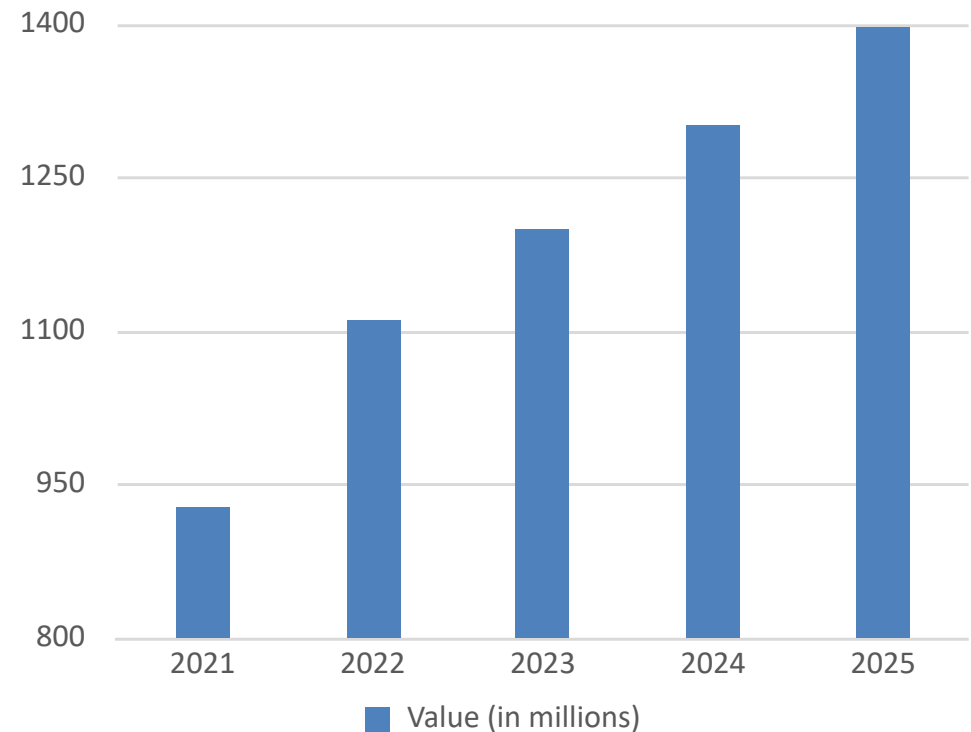
Economic Impact:

Growing attack frequency (+107% annually)

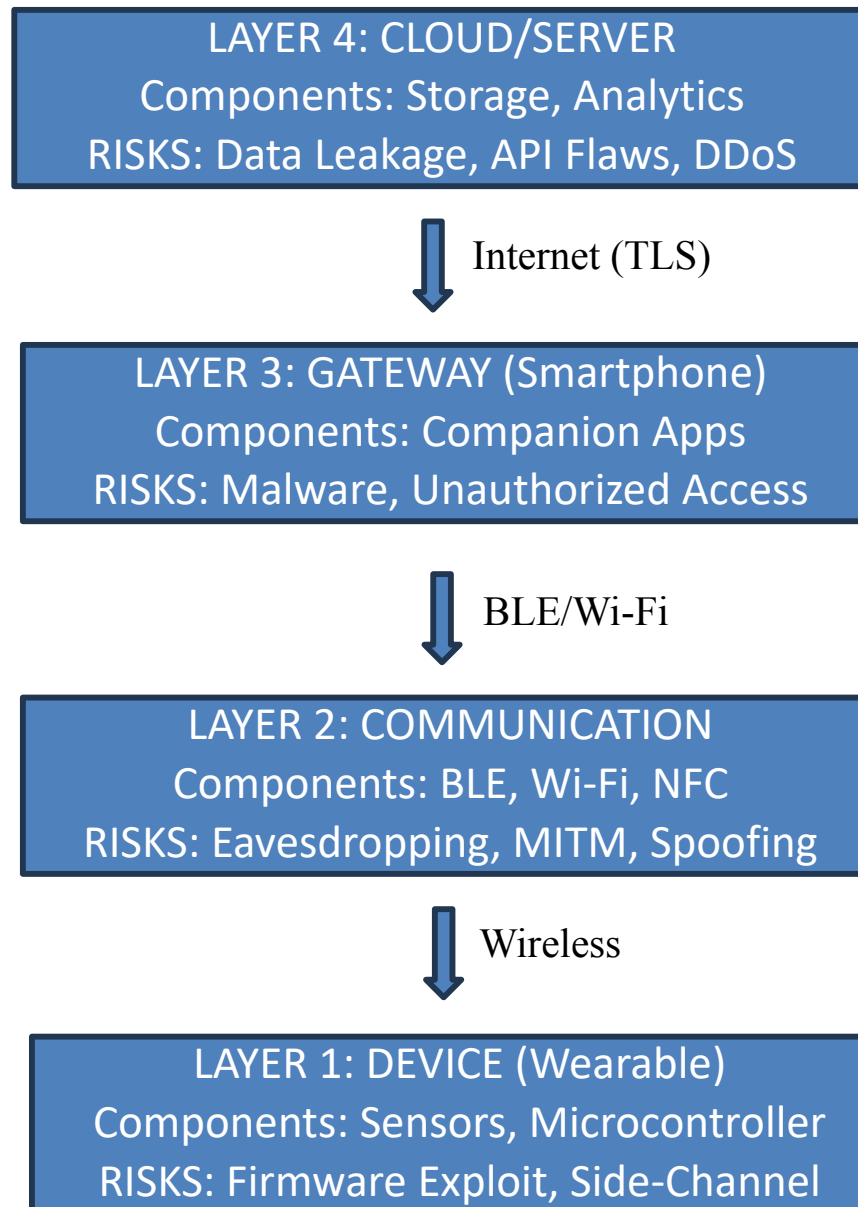
Rising breach costs

Trust erosion in wearable technology

Wearable Devices Growth



Which Layers Are at Risk?



Key Research Findings

Risk Assessment: $R = \Sigma[P(T) \times V \times I]$.

P(T) = Threat probability (**↑107%**)

V = Vulnerability level (**HIGH**)

I = Impact (**SEVERE** - health data)

SIX KEY FINDINGS:

1. Data Interception Most Prevalent

- BLE vulnerabilities widely exploited
- Traffic analysis reveals activities even with encryption

2. Weak Authentication Widespread

- Default credentials persist
- Lack of MFA in majority

3. Resource Constraints Limit Security

- Traditional encryption too heavy
- Battery vs. security trade-off

4. Lightweight Cryptography Effective

- PRESENT, SIMON, SPECK, Ascon work well
- ECC provides efficient asymmetric crypto

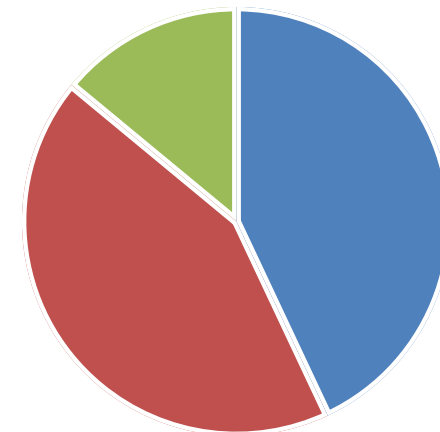
5. Firmware Security Neglected

- Manufacturers don't provide timely patches
- Hard-coded credentials common

6. Multi-Layered Defense Essential

- No single solution addresses all threats
- Comprehensive approach needed

Attack Severity



- MITM, Malware, Firmware
- Eavesdrop, DoS, Spoofing
- Side-Channel

Solutions Offered - Lightweight Cryptography

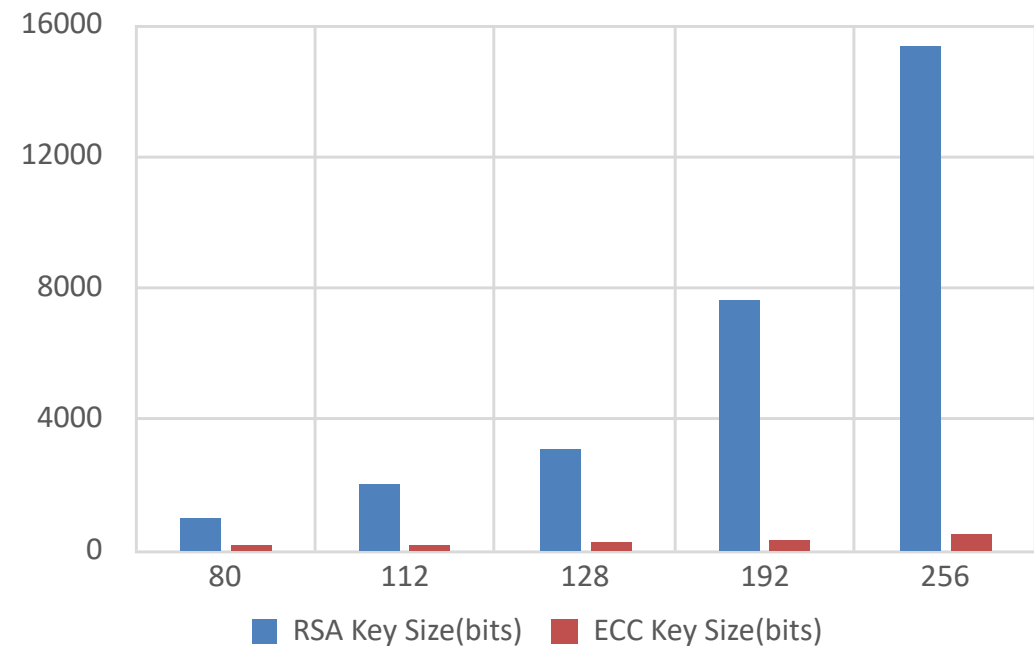
Why Lightweight Cryptography?

Lightweight cryptography is essential in wearable IoT devices because these devices operate under **severe resource constraints** and cannot support traditional heavy security mechanisms.

Algorithm	Key Size	Gate Equivalents	Efficiency
AES-128	128 bits	3,400 GE	Baseline (100%)
PRESENT	80/128 bits	1,570 GE	54% lighter
SIMON	64-128 bits	1,234 GE	64% lighter
SPECK	64-128 bits	984 GE	71% lighter
Ascon	128 bits	2,570 GE	NIST 2023

Table 2: Encryption Algorithm Comparison

RSA vs ECC



Authentication & Communication Solutions



AUTHENTICATION MECHANISMS:

Method	Security	Wearable Fit	Notes
Password/PIN	Low	⚠️ Weak	Default credentials exploited
Multi-Factor	High	✅ Good	99.9% breach reduction
Biometric	Very High	✅ Best	Continuous, hard to fake

TABLE 3: Authentication Methods

Biometric Authentication:

- Heart rate variability
- Gait patterns (walking style)
- Electrodermal activity
- **95-98% accuracy**
- Continuous implicit authentication

COMMUNICATION PROTOCOLS:

BLE Security Evolution:

BLE 4.0/4.1 (Legacy)

❌ Vulnerable to MITM

❌ Passive eavesdropping

↓ EVOLUTION

BLE 4.2+ (Secure Connections)

✅ ECDH key exchange

✅ MITM protection

Lightweight Cryptographic Algorithms:

- PRESENT, SIMON, SPECK, Ascon
- Designed for resource-constrained devices

Other Security Measures:

- **Elliptic Curve Cryptography (ECC):** Efficient public-key encryption
- **Multi-Factor Authentication (MFA):** Combining passwords with biometrics
- **Biometric Authentication:** Fingerprint, heart rate patterns
- **Secure Pairing Protocols:** Certificate-based authentication

Secure Communication Protocols:

- **BLE Secure Connections:** Bluetooth Low Energy security
- **CoAP with DTLS:** Constrained Application Protocol with security

Infrastructure Security:

- **Regular Firmware Updates:** Patching vulnerabilities promptly
- **Secure Boot Mechanisms:** Ensuring device integrity at startup
- **Network Segmentation:** Isolating wearable devices from critical systems

EMERGING TECHNOLOGIES:

Wearable IoT security is continuously evolving, and emerging technologies are playing a crucial role in addressing current limitations. The following innovations are expected to significantly enhance security, privacy, and resilience in wearable ecosystems:

1. Blockchain

- Decentralized authentication
- Immutable audit logs

2. AI & Machine Learning

- Real-time anomaly detection
- 98%+ threat accuracy

3. Secure Hardware

- TPM, PUF modules
- Hardware key protection

4. Post-Quantum Cryptography

- Future-proof security
- Quantum-resistant algorithms

The Challenge:

- 1.4B wearable devices by 2025
- 107% increase in attacks
- \$4.88M average breach cost

Key Takeaways:

- Wearable IoT devices face significant security challenges
- Lightweight security solutions are essential
- Multi-layered approach combining encryption, authentication, and secure protocols
- Emerging technologies (blockchain, AI, secure hardware) show promise

Future Outlook:

- Security is not optional—it's fundamental to user trust
- Continuous innovation needed to stay ahead of threats
- Industry-wide collaboration crucial for secure wearable IoT ecosystems

Thankyou